Aareen Satish Aher

Environmental Science & Engineering
Dual Degree (B.Tech. + M.Tech.)
Indian Institute of Technology Bombay

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Examination	University	Institute	Year	CPI
Graduation	IIT Bombay	IIT Bombay	2025	8.01

PROFESSIONAL & RESEARCH EXPERIENCES

University Of Alberta
Edmonton, Canada
[May'23-Jul'23]

*Selected as **sole** representative from IIT Bombay along with **3** applicants nationwide for prestigious UARE program*

- Optimized road-geometry impact assessment workflows by analysing traffic data rates using the **VISTA** simulator
- Increased data extraction accuracy by **15%** by developing pipeline that isolates pavement from **500+** road section
- Analysed **LIDAR** point cloud data from over **100+** scans with **Cloud Compare**, for reliable surface-level insights

Master's Thesis
ANT Lab, IIT Bombay
[Mar'24- May'25]

Machine Learning Model for Urban Particulate Matter Pollution Forecasting | Guide: Prof. Manoranjan Sahu

- Designed end-to-end $PM_{2.5}/PM_{10}$ time series forecasting pipeline across **3** monitoring stations for **24-72 hr** horizon forecast
- Implemented base model with **XGB, ANN, LSTM** on time series data achieving R^2 up to **0.94** under out-of-sample evaluation
- Implemented **5** hybrid ensemble method reducing forecast **RMSE** by **18-30%** and compressed error variance by **20-35%**
- Bayesian Model Averaging – XGB+LSTM achieved $PM_{2.5}$ **RMSE 1.43** $\mu g/m^3$ and PM_{10} RMSE **4.71** $\mu g/m^3$ across Mumbai

KEY PROJECTS

Chest CT Diagnosis System
Self-Project (Dec'25)

- Built **2-stage DenseNet121** pipeline **0.5k+** CT images achieving **99.6%** accuracy,**0.9** F1-score for binary detection
- Achieved **71.6%** accuracy with **0.72** weighted F1-score for **3** class carcinoma subtypes classification task result
- Implemented a **MLOPs** workflow with **DVC** for pipeline versioning & **MLflow** (via **Dagshub**) for experiment tracking
- Deployed Dockerized Flask inference on **AWS EC2**, with image stored in **AWS ECR** and automated **GitHub Actions CI/CD**, enabling **real-time** CT scan inference (**<1s**) while Integrating **Grad-CAM** explainability to visualize disease-relevant regions

Multimodal RAG PDF Assistant
Self-Project (Nov'25)

- Designed & built end-to-end multimodal **RAG** pipeline using **LangChain**, **Streamlit**, **Unstructured** & **ChromaDB**
- Ingested **10+** academic PDFs with title-aware chunking to enable structured text, tables and image retrieval
- Optimized semantic retrieval by embedding **1k+**docs with OpenAI, improving top-k relevance by **30%** via **MMR**
- Implemented grounded QA with **GPT-4o** using evidence-only prompt, reducing hallucination vs zero-context LLM
- Deployed **Streamlit** app with persistent vector store, multimodal evidence, achieving **<1.5s** average query latency

Corporate AI Assistant
Self-Project (Jan'26)

- Designed & deployed **enterprise-grade** RAG based **multi-user** AI platform using **n8n** and **OpenAI** embedding models
- Indexed 100+ internal policy docs into a semantic vector store, delivering HR/IT query resolution with **<1.2s** latency
- Implemented **LLM-based** intent classification, confidence scoring & rule-driven access-control gate for governed access
- Blocked **100%** unauthorized queries and improved response prediction to **0.89** F1 score across internal QA response

XAI-Constrained ML Optimization
Self-Project (Jan'26)

- Built an **explainability-constrained** ML optimization pipeline enabling credit risk modeling across **30+** features
- Optimized Random Forest & XGBoost through **XAI-aware** tuning, achieving **93.5%** accuracy, **0.95** ROC-AUC, **0.83** F1
- Enforced **model stability** using SHAP, PDP, ALE with **SHAP** variance decrease to **$2x10^{-3}$**, **0 PDP** curvature, **ALE < 0.41**
- Productionized ML experimentation with MLflow + DAGsHub, logging prams, metrics, and XAI artifacts for auditability

Predictive Churn Analytics
Self-Project (Jan'26)

- Architected end-to-end churn pipeline using **Microsoft SQL Server ETL** & **XGBoost** to automate **churn risk scoring**
- Designed staging and production workflow with validation logic and **SQL views** ensuring high integrity data
- Developed a **Power BI dashboard** providing service & account-level churn insight for executive decision-making
- Built and deployed ML **churn prediction** framework to identify high-risk customers and guide retention strategies

POSITIONS OF RESPONSIBILITY

Events Manager
Tatva | ESED

*Inaugural fest of Environmental Science & Engineering Department |Footfall: **500+**|Events & Workshop: **10+***

- Managed an INR **0.25M+** budget while leading overall strategy planning, execution & delivery of flagship events
- Boosted attendance by **40%** & revenue by **10%** by launching, scaling & leading campus-wide ambassador program

Department Placement Co.
Placement Team

*Assisting the placements of **1800+** students in the team of **64+** DPCs being the link between PT cell & students*

- Responsible for verifying **30+** resumes & conducting a strategic pitching deck for maximum student recruitment
- Key member of the **department council,** heading strategic initiatives & addressing concerns for over **45** students

TECHNICAL SKILLS

Language	C++, Python, SQL | **Libraries:** NumPy, Pandas, Matplotlib, Sckit-Learn, TensorFlow, OpenCV, LangChain, Chroma DB
MLOps	Docker, DVC, MLflow, GitHub CI/CD, Flask, n8n, Git, VS Code, Dagshub, Kubernetes |**Cloud Services:** AWS
Domain Tool	Unity, QGIS, EPANET, Visual MINTEQ, CloudCompare, OpenLCA, Google Earth Pro, BioWin, MATLAB, Power BI

EXTRACURRICULAR ACTIVITIES

Conference	• **Key Speaker** | LIDAR Application in Transport Engineering | Conference, University of Alberta | **150+** Scholar • Participant | AI, GenAI & ML Application | Amii Conference, Edmonton | Insight from **Richard Sutton** | **1k+**
Sports	• Competed **state-level** Aquathlon, showcasing high stamina, endurance & athleticism among **50+** competitor
Interest	• Travelling | Swimming | Listening Music | Web Series | Cinema & Concerts | Cricket | Football | Trekking